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EDUCATION	National University of Singapore , Singapore.	2025–now
	◦ Doctor of Philosophy, Computer Science.	Supervisor: Prof.HE Bingsheng
	National University of Singapore , Singapore.	2023–2025
	◦ Master of Computing, Computer Science Specialization.	GPA: 4.4/5.0
	South China University of Technology (SCUT) , Guangzhou, China.	2019–2023
	◦ B.Eng., Software Engineering.	GPA: 3.6/4.0

PRIZES AND AWARDS	• Excellent Degree Dissertation of South China University of Technology	2023
	• Honorable Mention in Mathematical Contest in Modeling	2023
	• National Scholarship	2022
	• Bronze Medal (46th) in ICPC Asia-East Continent Final(Xi An)	2022
	• 101/1608 in CCF-DBCI Competition of "Small Sample Data Classification"	2022
	• Silver Medal (46th) in ICPC Asia Regional Contest(Ji Nan)	2021
	• 44/3567 in CCF-DBCI Competition of "Recognition of figure skaters' skeleton points based on Paddle"	2021
	• First Prize in National Olympiad in Informatics in Province(NOIP)	2017

PREPRINTS AND PUBLICATIONS	• CHEN Han , Zining Zhang, Wenqi Pei, Bingsheng He, Ming Wu, Jason Zeng, Michael Heinrich, Wei Wu, Hongbao Zhang. MemForest: An Efficient Agent Memory System with Hierarchical Temporal Indexing . Revision for VLDB 2027	2026
	• CHEN Han , Tao Huang, Phuong Pham, Shuang Wu. HiAE: A High-Throughput Authenticated Encryption Algorithm for Cross-Platform Efficiency . Cryptology ePrint Archive, Report 2025/377	2025
	• CHEN Han , Zicong Jiang, Zining Zhang, Bingsheng He, Pingyi Luo, Mian Lu, Yuqiang Chen. LogQuant: Log-Distributed 2-Bit Quantization of KV Cache with Superior Accuracy Preservation . Accepted at ICLR 2025 Workshop on Sparsity in LLMs	2025
	• Wenqi Pei, Hailing Xu, Henry Hengyuan Zhao, Shizheng Hou, CHEN Han , Zining Zhang, Pingyi Luo, Bingsheng He. Feather-SQL: A Lightweight NL2SQL Framework with Dual-Model Collaboration Paradigm for Small Language Models . Published in IJCNLP-AAACL 2025 Findings; earlier version presented at ICLR 2025 DL4C Workshop as <i>Optimizing Small Language Models for NL2SQL</i>	2025
	• Wenqi Pei, Shizheng Hou, Boyan Li, CHEN Han , Zhichao Shi, Yuyu Luo. ROSE: An Intent-Centered Evaluation Metric for NL2SQL . Accepted at ACL 2026 Main	2026

- *Jian Yang, Zhizhuo Kou, Yao Tian, Hao Zhang, **CHEN Han**, Sirui Han, Yike Guo. **RW-TTT: Batched Serving for Request-Owned Test-Time Training State***. Submitted to EMNLP 2026 2026

PROJECT
EXPERIENCE

- ***Ph.D. Research: Efficient Agent Memory System for LLM Agents*** in **National University of Singapore**
Supervisor: [Prof. HE Bingsheng](#) Jan.–May 2026
 - Design *MemForest*, an agent memory system that reformulates LLM agent memory as a write-efficient temporal data problem, decoupling memory construction into parallelizable operations.
 - Propose *MemTree*, a hierarchical temporal index that organizes memory as time-ordered trees with localized per-node updates, replacing costly full-state rewrites.
 - Achieve 89% pass@8 accuracy on LongMemEval-S with approximately 6× higher memory construction throughput than state-of-the-art EverMemOS.
 - Sponsored by [0G Labs \(Zero Gravity Labs\)](#), a blockchain infrastructure company building the decentralized L1 for AI agents.
 - Revision for VLDB 2027. [\[Paper\]](#) [\[Code\]](#)
- ***Cryptography Engineer: Efficient Cipher Implementation and Whitebox Design*** in **SG Digital Trust Lab, Singapore Research Center, 2012 Laboratory**
Mentor: [Dr. WU Shuang](#) Jan.–June 2025
 - Design whitebox implementation of AES for different platforms, products and scenarios.
 - Implement HiAE cipher for kernel and user space, integrate it into several products, including mobile phone, server and auto-driving chips.
 - Optimize the performance of SM2, SM3 and SM4 ciphers on ARM platforms.
- ***Research Assistant: optimization for large language Model inference*** in **National University of Singapore**
Mentor: [Prof. HE Bingsheng](#) May–Sept. 2024
 - Design a new 2-bit KV Cache quantization for LLMs base on attention patterns. achieve over 200% accuracy improvement at same compression rate.
 - Implement a adaptive API for popular inference frameworks like Python’ s `transformers`, boosts batch size by 60% without increasing memory consumption.
 - Accepted at ICLR 2025 Workshop on Sparsity in LLMs. [\[Paper\]](#) [\[Code\]](#)
- ***Internship: Cryptography Engineer*** in **SG Digital Trust Lab, Singapore Research Center, 2012 Laboratory**
Mentor: [Dr. HUANG Tao](#) Jan.–Dec. 2024
 - Design a ‘XAXX’ structure to efficiently utilize the distinct pipelines of both ARM and x86 (with AES-NI) architectures, achieving high IPC.
 - Build a new AEAD (Authenticated Encryption with Associated Data) cipher named ‘HiAE’ based on the ‘XAXX’ structure, which is 5× faster than AES-256-GCM across different platforms and outperforms all existing AEAD ciphers on latest ARM and x86 processors.
 - Create a new record of 340Gbps throughput for single-threaded and single-stream AEAD encryption. Applied to the various products.
 - Open-source on Cryptology ePrint Archive [\[Paper\]](#) [\[Code\]](#)
- ***Research Assistant: Parallel Graph Partitioning*** in **Hong Kong University of Science and Technology, Guangzhou**
Mentor: [Prof. Zeyi WEN](#) Apr–Sept. 2023

- Implement a library of parallel hyper graph partitioning with openMP task.
- Realize a multiple node contraction algorithm for graph coursening.
- Research parallel graph partitioning algorithm and adapt it for fill-in reduction of sparse matrix.

- ***Symmetric Matrix Solving Algorithm Parallel Optimization for ARM Architecture***

Mentor: Prof. TANG Deyou

May-Dec. 2022

- Optimize and parallel Bounded Bunch-Kaufman Algorithm(*sysv_rk subroutine of LAPACK) for solving symmetric matrix on ARM server processor with NEON instruction set and openMP.
- Implement a parallel column reordering method in row swap of solving symmetric matrix to enhance memory access locality for column major matrix for better cache hit rate and parallelism, achieving a performance improvement from 320Gflops to 580Gflops.
- Implement the same optimization on Skylake Intel processor and achieve 2-5x multi-core speedup than MKL library for *sytrs_3 subroutine of LAPACK.
- Awarded as the Excellent Degree Dissertation of South China University of Technology.

TECHNICAL SKILLS

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- *English*: IELTS(6.5), CET-4, CET-6.
 - *Programming Languages*: C/C++, Fortran, p4-16, Python, SQL, L^AT_EX.
 - *Technical Skills*: openMP, SIMDs(NEON, AVX512), MPI, PyTorch, CUDA.
 - *TestDemo Certificate*: C++, TOP 10%, LINUX, TOP 10%, PYTHON, TOP 10%.
 - *Kaggle Certificate*: Data Visualization, Intro to Machine Learning, Intro to Deep Learning, Intro to Game AI and Reinforcement Learning.
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EXCHANGE EXPERIENCE

- **Online Academic Program on Machine Learning, McGill University** Jan.-Feb. 2022

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Google Scholar:
<https://scholar.google.com/citations?user=ru10E6wAAAAJ>
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教育经历

新加坡国立大学, 新加坡	2025–现在
◦ 计算机科学博士.	导师: 何丙胜教授
新加坡国立大学, 新加坡	2023–2025
◦ 计算机科学硕士.	GPA: 4.4/5.0
华南理工大学, 广东省广州市	2019–2023
◦ 工学学士, 软件工程专业.	GPA: 3.6/4.0

获奖荣誉

• 华南理工大学本科优秀毕业设计 (论文)	2023
• 二等奖美国大学生数学建模竞赛 (MCM/ICM)	2023
• 铜牌第 46 届 ICPC 国际大学生程序设计竞赛亚洲区决赛	2022
• 101/1608 CCF-DBCI ”小样本数据分类算法”竞赛	2022
• 国家奖学金	2022
• 银牌第 46 届 ICPC 国际大学生程序设计竞赛 (济南站)	2021
• 44/3567 CCF-DBCI ”基于飞浆实现花样滑冰选手骨骼点识别”竞赛	2021
• 一等奖全国青少年信息学奥林匹克联赛 (NOIP)	2017

预印本和发表
论文

• <i>CHEN Han, Zining Zhang, Wenqi Pei, Bingsheng He, Ming Wu, Jason Zeng, Michael Heinrich, Wei Wu, Hongbao Zhang. MemForest: An Efficient Agent Memory System with Hierarchical Temporal Indexing.</i> VLDB 2027 Revision	2026
• <i>CHEN Han, Tao Huang, Phuong Pham, Shuang Wu. HiAE: A High-Throughput Authenticated Encryption Algorithm for Cross-Platform Efficiency.</i> Cryptology ePrint Archive, Report 2025/377	2025
• <i>CHEN Han, Zicong Jiang, Zining Zhang, Bingsheng He, Pingyi Luo, Mian Lu, Yuqiang Chen. LogQuant: Log-Distributed 2-Bit Quantization of KV Cache with Superior Accuracy Preservation.</i> 录用于 ICLR 2025 Sparsity in LLMs Workshop	2025
• <i>Wenqi Pei, Hailing Xu, Henry Hengyuan Zhao, Shizheng Hou, CHEN Han, Zining Zhang, Pingyi Luo, Bingsheng He. Feather-SQL: A Lightweight NL2SQL Framework with Dual-Model Collaboration Paradigm for Small Language Models.</i> 发表于 IJCNLP-AAACL 2025 Findings; 早期版本以 <i>Optimizing Small Language Models for NL2SQL</i> 为题发表于 ICLR 2025 DL4C Workshop	2025
• <i>Wenqi Pei, Shizheng Hou, Boyan Li, CHEN Han, Zhichao Shi, Yuyu Luo. ROSE: An Intent-Centered Evaluation Metric for NL2SQL.</i> 录用于 ACL 2026 Main	2026
• <i>Jian Yang, Zhizhuo Kou, Yao Tian, Hao Zhang, CHEN Han, Sirui Han, Yike Guo. RW-TTT: Batched Serving for Request-Owned Test-Time Training State.</i> 已投稿至 EMNLP 2026	2026

- **博士研究：面向大语言模型智能体的高效记忆系统：**新加坡国立大学 2026/01–2026/05
导师：何丙胜教授
 - 设计了 *MemForest* 智能体记忆系统，将 LLM 智能体记忆重新建模为写入高效的时序数据问题，将记忆构建解耦为可并行操作。
 - 提出 *MemTree* 层次化时序索引结构，将记忆组织为时间有序的树，用局部节点更新替代代价高昂的全状态重写。
 - 在 LongMemEval-S 基准上达到 89% pass@8 准确率，同时记忆构建吞吐量约为最新方法 EverMemOS 的 6 倍。
 - 该项目由 0G Labs (Zero Gravity Labs) 赞助，一家为 AI 智能体构建去中心化 L1 区块链基础设施的公司。
 - VLDB 2027 Revision. [Paper] [Code]
- **密码算法工程师：高效密码算法实现和白盒设计：**新加坡数字信任实验室，华为 2012 实验室新加坡研究所 2025/01–2025/06
导师：吴双博士
 - 设计了不同场景和平台的 AES 白盒实现。
 - 在用户态和内核态实现了 HiAE 密码算法，并集成到多种产品中，包括手机、服务器和自动驾驶芯片。
 - 优化了 SM2、SM3 和 SM4 密码算法在 ARM 平台上的性能。
- **科研助理：大语言模型推理优化：**新加坡国立大学 2024/05–2024/09
导师：何丙胜教授
 - 设计了一种基于注意力模式的 2 位 KV Cache 量化方法，在相同压缩率下，提高了 200% 的准确率。
 - 实现了一个适应性 API，用于流行的推理框架，如 Python 的 transformers，在不增加内存消耗的情况下，将批处理大小提高了 60%。
 - 该项目已被 ICLR 2025 Sparsity in LLMs Workshop 接受。[Paper] [Code]
- **实习生：密码算法工程师：**华为 2012 实验室新加坡研究所谢尔德实验室 2024/01–2024/12
导师：黄涛博士
 - 设计了一种新的 'XAXX' 算法结构，可以高效利用 ARM 和 x86(带 AES-NI) 架构的流水线，实现了高 IPC。
 - 基于 'XAXX' 结构构建了一种新的 AEAD(带关联数据的认证加密) 密码算法，'HiAE'，在多种平台上相较 AES-256-GCM 提升 5 倍以上性能，在最新的 ARM 和 x86 处理器上优于所有现有的 AEAD 密码算法。
 - 创造了单线程单流 AEAD 加密的 340Gbps 新纪录，并应用于多种华为产品。
 - 该项目已在 Cryptology ePrint Archive 上开源。[Paper] [Code]
- **科研助理：香港科技大学广州校区** 2023/04–2023/09
导师：文泽忆教授
 - 实现了一个基于 openMP Task 的并行多层拓扑图分割库。
 - 在图压缩中实现了一种多节点收缩算法。
 - 研究将并行图分割算法，应用于稀疏矩阵的填充减少。
- **对称矩阵函数求解 BBK 算法的并行优化** 2022/04–2022/12
导师：汤德佑教授
 - 在 ARM 处理器上利用 NEON 指令集和 openMP 对 Bounded Bunch-Kaufman 算法 (LAPACK 库 *sysv_rk 函数) 进行并行优化。
 - 实现了一种并行列重排方法，在列优先矩阵的行交换中改进访存局部性，使得缓存命中率和并行性能得到提高，在鲲鹏 920-6426 处理器上的单精度性能从 320Gflops 提升到 580Gflops。

- 将该方法移植到 Intel Skylake 处理器上, 对比 MKL 库的 `*sytrs_3` 函数, 实现了 2-5 倍的并行性能提升。
- 该项目获评华南理工大学本科优秀毕业设计。

专业技能

- 英语认证水平: CET-4, CET-6, IELTS(6.5).
- 编程语言: C/C++, Fortran, p4-16, Python, SQL, $\LaTeX$.
- 编程技能: openMP, SIMDs(NEON, AVX512), MPI, PyTorch, CUDA.
- *TestDemo* 编程技能认证: C++, TOP 10%, LINUX, TOP 10%, PYTHON, TOP 10%
- *Kaggle* 课程认证: 数据可视化, 机器学习, 深度学习, 强化学习

交换经历

- 机器学习线上访学项目, 麦吉尔大学 2022/01–2022/02